PhD Committee Meeting Preparation Document

Comprehensive Talking Points, Facts, and Q&A

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1 Slide-by-Slide Detailed Content

1.1 Slide 1: Title

Opening Statement (30 seconds):

"Good morning/afternoon. Thank you for taking the time to attend my PhD committee review meeting. My research focuses on developing a Learning Automata framework for adaptive kelp forest monitoring in British Columbia coastal waters. This work addresses a critical environmental monitoring challenge using principled machine learning techniques."

Context to Set:

- This is a **committee review**, not the proposal defense
- Purpose: Get feedback on research direction and progress
- Supervisor: Dr. Neil Ernst (Computer Science)
- Committee: Dr. Maycira Costa (Geography - remote sensing), Dr. Kevin Stanley (Computer Science - ML)

1.2 Slide 2: Motivation

Key Statistics to Memorize:

Fact	Source
BC coastline length	26,000+ km
Some regions kelp loss	>70% (Starko et al. 2024)
Thermal threshold for kelp stress	18–20°C SST
Sentinel-2 resolution	10m spatial, 5-day revisit
Primary species in BC	Bull kelp (*Nereocystis luetkeana*)
Bull kelp life cycle	Annual (recruits spring, dies after reproduction)
Marine heatwave "Blob"	2014–2016, caused widespread mortality

Detailed Talking Points:

Why Kelp Matters:

- **Productivity**: Net primary production of 500–1,000+ g C m^{-2} $year^{-1}$ in dense stands—rivals tropical rainforests
- **Ecosystem services**: (1) habitat for commercially important fish (herring, rockfish); (2) carbon sequestration; (3) coastal protection via wave attenuation; (4) nutrient cycling
- **Cultural significance**: Coastal First Nations have sustained traditional kelp harvesting practices for millennia. Archaeological evidence shows human-kelp interactions spanning thousands of years.
- **Species**: Bull kelp (*Nereocystis luetkeana*) dominates BC. It's an **annual** species (unlike California's perennial giant kelp), making it especially sensitive to conditions during spring-summer growth.

The Problem - Regional Variation:

- **Southern BC (thermal stress)**: Summer SST regularly exceeds 18–20°C threshold. Berry et al. (2021) documented 63% decline in South Puget Sound over 145 years.
- **Northern BC (trophic cascades)**: Sea star wasting disease (2013–2017) eliminated sunflower stars (*Pycnopodia helianthoides*), causing urchin population explosion and "urchin barrens."
- **Key insight**: The **causes of decline are spatially heterogeneous**, meaning detection methods need to adapt to local conditions.

Monitoring Challenge:

- **Spectral variability**: Kelp signature changes with water turbidity, sun glint, atmospheric conditions, tide height, and kelp physiological state.
- **Algorithm failure modes**: Methods working under ideal conditions (calm water, low tide, minimal cloud) fail when conditions deviate.
- **Label scarcity**: Ground-truth surveys are expensive, logistically challenging in remote areas, and cannot keep pace with satellite acquisition.

1.3 Slide 3: Problem Statement

Core Technical Challenge:

The fundamental tension is between:

1. **Need for site-specific optimization**: Different environments need different detection strategies
2. **Difficulty obtaining labeled data**: Cannot get sufficient training data for every environmental context

Four Interconnected Problems:

1. **Environmental heterogeneity**:
   - Mora-Soto et al. (2024) identified **4 distinct environmental clusters** in southern Salish Sea alone
   - Each cluster shows different kelp growth patterns and spectral characteristics
   - Bull kelp in exposed outer coast behaves differently from protected inland waters
2. **Limited labeled data**:
   - Deep learning requires 1000s–10000s of labeled examples
   - Ground truth comes from: ShoreZone photography, WorldView classifications, in-situ surveys, citizen science
   - Labels arrive **irregularly and with delay** (days to months after imagery)
3. **Sparse and delayed feedback**:
   - Traditional supervised learning assumes dense, contemporaneous labels
   - In operational monitoring, you might get validation for 1–10% of predictions

- Delay can be 1 day to 6 months

4. **TEK integration**:
   - Indigenous communities have observed kelp dynamics for millennia
   - Integration must respect data sovereignty and community protocols

Why Learning Automata?

- LA provides mathematically grounded framework for learning optimal actions from sparse feedback
- Unlike supervised learning, LA only needs sparse reward signals to shift probability mass
- The "inaction" property: missing labels leave probabilities unchanged (graceful handling of sparse feedback)
- Proven convergence guarantees (ϵ-optimal)

1.4 Slide 4: Research Questions

RQ1 (Adaptation with Limited Labels):

- **Full question**: "How can we adapt kelp segmentation to diverse environmental conditions without extensive labeled data per region?"
- **Hypothesis**: LA can achieve effective adaptation using **10–100× fewer labels** than supervised approaches
- **Mechanism**: $L_{R\text{-}I}$ only needs sparse binary/continuous feedback to shift probability mass toward better-performing heuristics
- **Validation**: Compare label efficiency curves (performance vs. number of labels) between LA, supervised CNN, online learning baseline

RQ2 (Heterogeneous Region Coverage):

- **Full question**: "Can adaptive selection of detection methods enable effective monitoring across heterogeneous regions where single models fail?"
- **Hypothesis**: Different heuristics have **complementary failure modes**:
  - Spectral indices: fail under turbid conditions, robust to atmospheric variation
  - Deep learning: capture complex patterns, can overfit to training distribution
- **Validation**: Stratify test set by environmental conditions (turbidity quartiles, SST ranges), compare LA vs. best single heuristic vs. oracle

RQ3 (Sparse Label Learning):

- **Full question**: "How can a monitoring system improve its performance during deployment using sparse, asynchronously-arriving validation labels?"
- **Hypothesis**: Buffering + delayed reward propagation enables effective learning despite asynchronous feedback

- **Mechanism**: Temporal decay $\gamma^{\Delta t}$ downweights stale feedback that may no longer reflect current conditions
- **Validation**: Simulate different delay distributions (days/weeks/months), compare LA vs. baseline discarding delayed feedback vs. baseline treating delayed as immediate

RQ4 (TEK Integration):

- **Full question**: "How can Traditional Ecological Knowledge from coastal First Nations be integrated into automated kelp forest monitoring?"
- **Hypothesis**: TEK provides complementary temporal depth, spatial resolution, and contextual understanding
- **Integration points**: (1) Site importance weights w_s for prioritizing culturally significant areas; (2) temporal priors from traditional calendars; (3) elder validation of ambiguous detections
- **Key principle**: Two-Eyed Seeing (Etuaptmumk) - drawing on strengths of both indigenous and Western knowledge systems

1.5 Slide 5: Learning Automata Background

Mathematical Framework to Know:

Environment $E = \{\alpha, c, \beta\}$:

- $\alpha = \{\alpha_1, \alpha_2, \ldots, \alpha_r\}$: Set of r input actions (our 12 heuristics)
- β: Feedback response. Three models:
  - **P-model**: Binary $\beta \in \{0, 1\}$ (reward/penalty)
  - **Q-model**: Finite discrete values in $[0, 1]$
  - **S-model**: Continuous random variable in $[0, 1]$ — **we use this** (IoU, F1 are continuous)
- $c = \{c_1, c_2, \ldots, c_r\}$: Penalty probabilities $c_i = P(\beta = 1|\alpha_i)$

Automaton $A = \{\Phi, \alpha, \beta, F, G\}$:

- Φ: Internal states
- F: State transition function $\phi(n + 1) = F[\phi(n), \beta(n)]$
- G: Output mapping (action selection)

$\mathbf{L_{R\text{-}I}}$ Update Equations:

When action α_i selected and receives reward ($\beta = 0$):

$$p_i(n + 1) = p_i(n) + \lambda[1 - p_i(n)] \tag{1}$$

$$p_j(n + 1) = (1 - \lambda)p_j(n), \quad j \neq i \tag{2}$$

When penalty received ($\beta = 1$): **No change** (inaction)

Key Properties:

- $\lambda \in (0, 1)$: Learning rate

- **ϵ-optimal**: Probability of converging to optimal action approaches 1 as $\lambda \to 0$
- **Absorbing Markov chain**: Converges to one of r absorbing states (unit vectors)

For S-model (our case):

$$p_i(n+1) = p_i(n) + \lambda \cdot R \cdot [1 - p_i(n)] \tag{3}$$
$$p_j(n+1) = p_j(n) - \lambda \cdot R \cdot p_j(n), \quad j \neq i \tag{4}$$

where $R \in [0, 1]$ is the continuous reward (IoU score).

Non-Stationary Environments:

- **MSE (Markovian Switching)**: Environment states form a Markov chain. Models **inter-annual variability** (unpredictable regime shifts).
- **PSE (Periodic Switching)**: Environment cycles through k states every T events. Models **seasonal dynamics** (kelp phenology).

Historical Context:

- Originated: Soviet Union, 1960s (Tsetlin)
- My prior work: Object Migration Automata (OMA) for adaptive data structures (Bisong & Oommen 2019–2021)
- Applications: Image segmentation, classifier selection, sensor placement, wireless networks

1.6 Slide 6: Framework Architecture

High-Level Flow:

1. **Input**: Satellite tile $\mathbf{x}_t \in \mathbb{R}^{B \times H \times W}$ where $B = 5$ bands, $H \times W = 256 \times 256$ pixels
2. **Context extraction**: $\mathbf{c}_t \in \mathbb{R}^D$ encoding environmental conditions
3. **Heuristic selection**: LA samples heuristic h_i from probability vector $\mathbf{p}$
4. **Prediction**: Apply heuristic to get $\hat{\mathbf{y}}_t \in \{0, 1\}^{H \times W}$
5. **Buffering**: Store prediction until validation label arrives
6. **Update**: When label $\mathbf{y}_t$ arrives, compute reward $R = \text{IoU}(\hat{\mathbf{y}}_t, \mathbf{y}_t)$, update LA

Environment-Automaton Interaction:

- **Environment**: The kelp detection problem instance (tile + context)
- **Action**: Selected heuristic from pool
- **Response**: Delayed feedback $\beta(n) \in [0, 1]$ (IoU score when label arrives)

Why This Architecture:

- Decouples learning (LA) from prediction (heuristics)
- Heuristics can be pre-trained, frozen during deployment
- LA adapts the **selection policy**, not the heuristics themselves
- Handles any heuristic that produces binary segmentation mask

1.7 Slide 7: Framework Components

Component 1: Global Policy Learning

- **What it does**: Maintains probability vector $\mathbf{p} = [p_1, \ldots, p_{12}]$ over 12 heuristics
- **Initialization**: Uniform $p_i(0) = 1/12$
- **Update**: $\mathrm{L_{R\text{-}I}}$ with S-model rewards (IoU scores)
- **Key insight**: This is the "global" policy that applies everywhere. It learns which heuristics are generally better.
- **Multi-objective extension**:

$$R_{\text{total}} = w_1 \cdot R_{\text{IoU}} + w_2 \cdot R_{\text{speed}} + w_3 \cdot R_{\text{uncertainty}} + w_4 \cdot R_{\text{carbon}}$$

Heuristic Pool (12 methods in 3 tiers):
Tier 1 - Spectral Indices (4):

1. NDVI threshold: $(B_{\text{NIR}} - B_{\text{Red}})/(B_{\text{NIR}} + B_{\text{Red}}) > \tau$
2. FAI threshold: Floating Algae Index using red-edge bands
3. GNDVI threshold: Green NDVI (green instead of red)
4. Multi-index ensemble: Majority vote of above 3

Tier 2 - Classical ML (4):

5. Random Forest: 100-tree ensemble on spectral features
6. Gradient Boosting: XGBoost classifier
7. SVM with RBF kernel
8. K-means clustering + class assignment

Tier 3 - Deep Learning (4):

9. U-Net (basic): ResNet-18 encoder
10. U-Net (deep): ResNet-50 encoder
11. DeepLab v3+: Atrous spatial pyramid pooling
12. Lightweight U-Net: MobileNet encoder (fast inference)

Component 2: Context-Aware Local Adaptation

- **What it does**: Enables generalization across heterogeneous conditions by learning context-dependent policies
- **Based on**: Generalized Learning Automata (GLA) principles

Context Feature Vector $\mathbf{c} \in \mathbb{R}^D$:

- **Spectral**: Mean, variance of each band
- **Spatial**: Texture features (entropy, homogeneity)
- **Temporal**: Day of year, time since last valid observation
- **Environmental**: SST (from NOAA CoralTemp), tide height (CHS predictions), solar zenith angle
- **Geographic**: Latitude, longitude, distance to shore

Memory Bank + k-NN Retrieval:

- Memory bank $\mathcal{M} = \{(\mathbf{c}_j, \mathbf{p}_j, R_j)\}_{j=1}^{M}$
- Use FAISS for efficient approximate nearest neighbor search
- Retrieve k most similar contexts: $\mathcal{N}_k(\mathbf{c}) = \text{arg-}k\text{-min}_{\mathbf{c}_j} \|\mathbf{c} - \mathbf{c}_j\|_2$

Local Probability Estimation:

$$\mathbf{p}_{\text{local}}(\mathbf{c}) = \frac{\sum_{j \in \mathcal{N}_k(\mathbf{c})} w_j \cdot R_j \cdot \mathbf{p}_j}{\sum_{j \in \mathcal{N}_k(\mathbf{c})} w_j \cdot R_j}$$

where $w_j = \exp(-\|\mathbf{c} - \mathbf{c}_j\|_2^2/\sigma^2)$

Global-Local Blending:

$$\mathbf{p}_{\text{final}}(\mathbf{c}) = \alpha(\mathbf{c}) \cdot \mathbf{p}_{\text{global}} + (1 - \alpha(\mathbf{c})) \cdot \mathbf{p}_{\text{local}}(\mathbf{c})$$

- $\alpha \to 1$ when few similar entries exist (favor global)
- $\alpha \to 0$ when many similar entries with consistent performance (favor local)

Component 3: Sparse Label Handling

- **What it does**: Handles asynchronous arrival of validation labels

Prediction Buffer:

$$\mathcal{B} = \{(t_j, \mathbf{x}_j, \mathbf{c}_j, i_j, \hat{\mathbf{y}}_j)\}$$

Stores: timestamp, input tile, context, selected heuristic index, prediction

Hash Map Lookup: $\mathcal{L} : (\text{location}, \text{time}) \to \text{buffer_index}$ for O(1) lookup

Label Arrival Processing:

1. Look up entry in buffer using $\mathcal{L}(\ell, t)$
2. Compute reward $R = \text{IoU}(\hat{\mathbf{y}}, \mathbf{y})$
3. Update Component 1 global policy with (i, R)
4. Update Component 2 memory bank with $(\mathbf{c}, \mathbf{p}_{\text{final}}, R)$
5. Remove entry from buffer

Buffer Management:

- Entries older than $T_{\max} = 6$ months removed without update
- Assumption: conditions changed too much for reward to be informative

Temporal Decay:

$$R_{\text{effective}} = R \cdot \gamma^{\delta}$$

where $\delta = t_{\text{label}} - t_{\text{prediction}}$ (delay in days), $\gamma < 1$ is decay factor (e.g., 0.99)

1.8 Slide 8: Experimental Design

Dataset Specifications:

Imagery - Sentinel-2 Level-2A:

- **Spatial resolution**: 10m (bands 2,3,4,8), 20m (band 5, resampled to 10m)
- **Temporal coverage**: 2017–2024 (8 years)
- **Revisit time**: 5 days (twin satellite constellation)
- **Tile size**: 256×256 pixels = 2.56 km^2
- **Bands used**: B2 (blue), B3 (green), B4 (red), B5 (red-edge), B8 (NIR)
- **Preprocessing**: Cloud/shadow masking (Scene Classification Layer), land masking (vector shoreline)

Ground Truth Labels:

- **ShoreZone**: Oblique photography interpretations (historical baseline, updated every 10–15 years)
- **WorldView**: High-resolution (1.2–2.6m) classifications where available. Schroeder (2020) achieved 86.9% accuracy in Cowichan Bay.
- **In-situ surveys**: SCUBA transects, shoreline surveys from research partners
- **Citizen science**: iNaturalist, kelp-specific projects

Environmental Context Data:

- **SST**: NOAA CoralTemp product
- **Tide height**: Canadian Hydrographic Service (CHS) tidal predictions
- **Ocean color**: Sentinel-3 OLCI (chlorophyll, turbidity)

Three Evaluation Protocols:

Protocol A - Temporal Adaptation:

- Train: 2017–2020 (4 years)
- Validate: 2021–2022 (2 years)
- Test: 2023–2024 (2 years)
- Tests adaptation to interannual variability and climate anomalies

Protocol B - Spatial Adaptation:

- Leave-one-region-out cross-validation
- Regions: Southern Vancouver Island, Central Coast, Northern Coast, Haida Gwaii
- Tests generalization along geographic and environmental gradients

Protocol C - Sparse Label Simulation:

- Withhold varying fractions of labels: 10%, 50%, 90%
- Introduce variable delay: 1 day to 6 months (uniform, exponential distributions)
- Measures learning efficiency as function of label availability

Metrics:
Detection Performance:

- IoU (Intersection over Union): $\frac{|P \cap G|}{|P \cup G|}$
- F1-score: $\frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
- Precision: $\frac{TP}{TP+FP}$
- Recall: $\frac{TP}{TP+FN}$
- AUC (for probabilistic outputs)

Adaptation Metrics:

- **Regret**: Cumulative difference from oracle that always selects best heuristic
- **Convergence rate**: Number of labels needed to achieve 90% of asymptotic performance
- **Stability**: Variance in heuristic selection probabilities over time

Computational Efficiency:

- Inference time per tile
- Memory footprint
- Carbon cost (CodeCarbon)

Ablation Studies:

1. Without Component 2: Global policy only (no context-aware adaptation)
2. Without Component 3: Assume immediate labels (no delayed reward handling)
3. Without pursuit/estimator extensions: Basic $L_{\text{R-I}}$ only
4. Heuristic pool ablations: Remove each tier to assess contribution
5. Context feature ablations: Subset of context features to identify most informative

1.9 Slide 9: Expected Contributions

Scientific Contributions:

1. **Novel LA application to environmental monitoring**: First application to satellite-based ecosystem monitoring. Prior LA applications: image segmentation (Cuevas 2011), sensor placement (Ben-Zvi 2018), wireless networks (Nicopolitidis 2011).

2. **Empirical characterization of heuristic complementarity**: Will provide evidence for/against hypothesis that diverse heuristic pools outperform specialized approaches.

3. **Understanding sparse feedback learning dynamics**: Characterize how learning efficiency degrades with label scarcity and delay.

Methodological Contributions:

1. **Sparse label handling mechanism**: Buffering + delayed reward propagation. General enough for other domains with asynchronous feedback.

2. **Context-aware adaptation framework**: Memory-based similarity matching + global-local blending. Template for systems that must generalize across heterogeneous conditions.

3. **Multi-objective reward formulation**: Accuracy + speed + uncertainty + carbon. Addresses environmental cost of ML.

4. **Evaluation protocols**: Temporal, spatial, sparse label protocols as benchmarks for future adaptive monitoring studies.

5. **Multi-Source Environmental Data Fusion Framework (KEY CONTRIBUTION - SEPARATE PAPER):** Systematic methodology for fusing heterogeneous data sources (Sentinel-2 at 10m, SST at 5km, Sentinel-3 at 300m, tide stations, bathymetry) with principled handling of resolution mismatches, temporal alignment, and missing data. This is a **standalone methodological contribution** applicable to any coastal ecosystem monitoring application.

Practical Applications:

- Coastline-scale kelp extent maps (weekly to monthly updates)
- Early warning of kelp forest decline
- Monitoring recovery after disturbance
- Prioritization for field survey or intervention
- Blue carbon accounting
- Framework transferable to seagrass, coral reefs, terrestrial vegetation

REVISED PUBLICATION PLAN (5 Papers):

1. **Paper 1 (RQ1):** Core LA Framework for Adaptive Kelp Detection
2. **Paper 2 (RQ2):** Context-Aware Heuristic Selection Across Heterogeneous Environments
3. **Paper 3 (RQ3):** Learning from Sparse and Delayed Labels in Environmental Monitoring
4. **Paper 4 (NEW - Data Fusion):** Multi-Source Environmental Data Fusion for Coastal Monitoring: Handling Resolution Mismatches and Missing Data
5. **Paper 5 (RQ4):** TEK Integration Framework for Kelp Forest Monitoring

1.10 Slide 10: Summary

Key Takeaways (30-second version):

"My research addresses the challenge of adaptive kelp monitoring with limited labels and heterogeneous conditions. The solution is a Learning Automata framework with three components: global policy learning using $L_{R\text{-}I}$, context-aware local adaptation using memory-based similarity matching, and sparse label handling with temporal decay. I will validate this through four research questions with rigorous experimental design across temporal, spatial, and sparse label dimensions. The target is October 2027 completion with **five publications**. The impact is scalable monitoring supporting kelp conservation across BC's 26,000+ km coastline."

2 Dataset Usage in the LA Framework - Complete Details

This section provides a comprehensive explanation of exactly how each data element flows through the Learning Automata framework. This is critical for the committee to understand the concrete data requirements and processing pipeline.

2.1 Overview: Three Data Streams

The framework requires three distinct data streams, each serving a specific purpose:

Data Stream	Source	Role in Framework
Satellite Imagery	Sentinel-2 Level-2A	Input to heuristics for kelp prediction
Environmental Context	Multiple sources (NOAA, CHS, derived)	Determines which heuristic to select
Ground Truth Labels	Surveys, WorldView, citizen science	Provides reward signal for LA learning

2.2 Data Stream 1: Satellite Imagery (Input to Heuristics)

What it is: Sentinel-2 multispectral satellite imagery covering BC coastal waters.

Source: European Space Agency (ESA) Copernicus program, freely available via:

- Google Earth Engine (GEE) - primary access method
- Copernicus Open Access Hub - alternative
- AWS Open Data Registry - for bulk download

Specifications:

Parameter	Value
Product Level	Level-2A (surface reflectance, atmospherically corrected)
Spatial Resolution	10m (B2, B3, B4, B8), 20m (B5, resampled to 10m)
Temporal Resolution	5 days (Sentinel-2A + 2B constellation)
Spectral Bands Used	B2 (490nm blue), B3 (560nm green), B4 (665nm red), B5 (705nm red-edge), B8 (842nm NIR)
Tile Size	256 × 256 pixels = 2.56 km × 2.56 km
Archive Period	2017–2024 (8 years)
Geographic Extent	BC coastal zone (approximately 1 km inland to 5 km offshore)

Preprocessing Pipeline:

1. **Cloud/Shadow Masking**: Use Scene Classification Layer (SCL) from Level-2A product
   - Remove pixels classified as: cloud (high/medium/low probability), cloud shadow, cirrus
   - Retain only: water, vegetation, not vegetated, unclassified
2. **Land Masking**: Apply vector shoreline from BC Geographic Warehouse
   - Buffer 100m inland to capture intertidal zone
   - Purpose: Focus on water pixels where kelp can occur
3. **Quality Filtering**: Reject tiles with >30% masked pixels
4. **Tiling**: Extract 256×256 pixel tiles along coastline
   - 50% overlap between adjacent tiles for continuity
   - Each tile tagged with: center lat/lon, acquisition timestamp, tile ID
5. **Normalization**: Scale reflectance values to [0, 1] range

How imagery enters the framework:

1. Preprocessed tile $\mathbf{x}_t \in \mathbb{R}^{5\times256\times256}$ arrives with metadata (timestamp t, location ℓ)
2. Tile is passed to the **selected heuristic** h_i for prediction
3. Heuristic outputs binary mask $\hat{\mathbf{y}}_t \in \{0,1\}^{256\times256}$
4. The LA does NOT see the raw imagery—only the heuristics process pixels

Data Volume Estimates:

- BC coastline: 26,000 km
- Coastal zone tiles: ∼4,000 tiles per acquisition

- Acquisitions per year: ∼73 (5-day revisit)
- Usable fraction (after cloud filtering): ∼30%
- Usable tiles per year: ∼88,000
- 8-year archive: ∼700,000 tile-observations
- Storage: ∼50 GB (compressed) for all bands

2.3 Data Stream 2: Environmental Context Features

Purpose: Context features enable the LA to learn *which heuristic works best under which conditions*. Without context, the LA can only learn a single global policy. With context, it can adapt to local conditions.

The key insight: A heuristic that works well in clear water may fail in turbid water. By encoding environmental conditions as features, the memory bank can store "in turbid conditions, heuristic X worked well" and retrieve this knowledge when similar conditions occur.

2.3.1 Context Feature Categories and Sources

Category 1: Spectral Features (Derived from Imagery)

These are computed directly from the satellite tile $\mathbf{x}_t$:

Feature	Computation	Why It Matters
Mean reflectance (per band)	$\mu_b = \frac{1}{N}\sum_i x_{b,i}$ for each band b	Indicates water clarity, atmospheric conditions
Variance reflectance (per band)	$\sigma_b^2 = \frac{1}{N}\sum_i (x_{b,i} - \mu_b)^2$	High variance = heterogeneous scene (mixed kelp/water)
NIR/Red ratio	$\mu_{\text{NIR}}/\mu_{\text{Red}}$	Proxy for vegetation presence
Blue/Green ratio	$\mu_{\text{Blue}}/\mu_{\text{Green}}$	Water turbidity indicator

Total spectral features: 5 means + 5 variances + 2 ratios = **12 features**

Category 2: Spatial Texture Features (Derived from Imagery)

Computed using Gray-Level Co-occurrence Matrix (GLCM) on the NIR band:

Feature	Definition	Why It Matters
Entropy	$-\sum_{i,j} P_{i,j} \log P_{i,j}$	High entropy = complex texture (kelp canopy)
Homogeneity	$\sum_{i,j} \frac{P_{i,j}}{1+\lvert i-j\rvert}$	High homogeneity = uniform (open water)
Contrast	$\sum_{i,j} (i-j)^2 P_{i,j}$	Edge strength indicator
Correlation	Normalized correlation of GLCM	Texture directionality

Total texture features: 4 features

Category 3: Temporal Features (Derived from Metadata)

Feature	Source	Why It Matters
Day of year (DOY)	Timestamp t	Kelp phenology is seasonal (peak in summer)
Days since last observation	$t - t_{\text{prev}}$ at same location	Data continuity indicator
Year	Timestamp t	Interannual variability (e.g., heatwave years)

Total temporal features: 3 features
Category 4: Environmental Features (External Data Sources)
These require auxiliary datasets matched to the tile's location and time:

Feature	Source & Access	Why It Matters
Sea Surface Temperature (SST)	NOAA CoralTemp (5km, daily). Access via ERDDAP API	Kelp stressed above 18–20°C
Tide Height	CHS Tidal Predictions. Access via Fisheries & Oceans API	Kelp visibility varies with tide
Chlorophyll-a	Sentinel-3 OLCI (300m, daily). Access via GEE	Proxy for water productivity
Turbidity (Kd490)	Sentinel-3 OLCI	Water clarity affects spectral signature
Solar Zenith Angle	Computed from lat/lon/time	Sun glint varies with sun angle

Total environmental features: 5 features
Category 5: Geographic Features (Derived from Location)

Feature	Source	Why It Matters
Latitude	Tile center coordinate	North-south environmental gradient
Longitude	Tile center coordinate	East-west exposure gradient
Distance to shore	Computed from shoreline vector	Kelp occurs near shore
Fetch (wave exposure)	Precomputed from bathymetry	Exposed vs. sheltered sites
Depth (bathymetry)	CHS bathymetry dataset	Kelp needs substrate within depth range

Total geographic features: 5 features

2.3.2 Complete Context Vector

Total dimension: $D = 12 + 4 + 3 + 5 + 5 = \mathbf{29}$ features

The context vector for tile $\mathbf{x}_t$ at location ℓ and time t is:

$$\mathbf{c}_t = [\underbrace{\mu_{B2}, \ldots, \sigma^2_{B8}, r_1, r_2}_{\text{12 spectral}}, \underbrace{H, \text{Hom}, C, \text{Cor}}_{\text{4 texture}}, \underbrace{\text{DOY}, \Delta t, Y}_{\text{3 temporal}}, \underbrace{\text{SST}, h_{\text{tide}}, \text{Chl}, K, \theta}_{\text{5 environmental}}, \underbrace{\phi, \lambda, d_s, F, z}_{\text{5 geographic}}]$$

2.3.3 Context Feature Acquisition Pipeline

When a new tile arrives, the following happens:

1. **Spectral/Texture Features** (computed on-the-fly):
   - Input: Raw tile $\mathbf{x}_t$
   - Computation: NumPy operations for statistics, scikit-image for GLCM
   - Time: <100 ms per tile
2. **Temporal Features** (from metadata):
   - Input: Timestamp t
   - Computation: Date parsing, lookup of previous observation time
   - Time: <1 ms
3. **Environmental Features** (from cached external data):
   - **Pre-download strategy**: Download SST, tide, ocean color for entire study period at start
   - **Storage**: NetCDF files indexed by date and lat/lon grid
   - **Lookup**: Bilinear interpolation to tile center location
   - Time: <10 ms per tile (from local cache)
4. **Geographic Features** (from precomputed layers):
   - **Pre-compute strategy**: Create raster layers for distance-to-shore, fetch, bathymetry
   - **Storage**: GeoTIFF files covering BC coastal zone
   - **Lookup**: Direct pixel lookup based on tile center
   - Time: <5 ms per tile
5. **Normalization**:
   - Z-score normalization: $c'_i = (c_i - \mu_i)/\sigma_i$
   - μ_i, σ_i computed from training set statistics
   - Purpose: Ensure all features on similar scale for k-NN distance

2.3.4 Spatial and Temporal Alignment of External Data Sources

The Alignment Problem:

External data sources (SST, tide, ocean color, bathymetry) have different:

- **Spatial resolutions**: ranging from 10m to 25km
- **Temporal resolutions**: ranging from hourly to static
- **Coordinate systems**: some in lat/lon, some in projected coordinates
- **Grid structures**: regular grids vs. irregular station networks

We need to extract a single value for each feature at the tile's location and time.

Alignment Strategy by Data Source:

Data Source	Native Spatial Res.	Native Temporal Res.	Alignment Method	Details
NOAA CoralTemp SST	5 km	Daily	Bilinear interp.	Interpolate 4 nearest grid points to tile center
CHS Tide	Station-based	6-minute	Nearest station + time interp.	Find nearest tide station, interpolate to image time
Sentinel-3 OLCI (Chl, Kd490)	300 m	1–2 days	Bilinear interp. + temporal nearest	Spatially interpolate, use closest cloud-free date
Solar Zenith Angle	Computed	Instantaneous	Direct calculation	Compute from lat, lon, datetime using PyEphem
CHS Bathymetry	100 m	Static	Bilinear interp.	One-time precomputation for study area
Distance to Shore	10 m	Static	Direct lookup	Precomputed raster, exact pixel match
Fetch (wave exposure)	100 m	Static	Bilinear interp.	Precomputed from wind rose and coastline

Detailed Alignment Procedures:

1. SST Alignment (NOAA CoralTemp):

```
def get_sst(tile_lat, tile_lon, tile_datetime):
    # CoralTemp is on 0.05° grid (~5km)
    # Find the 4 surrounding grid points
    lat_grid = np.arange(-90, 90, 0.05)
    lon_grid = np.arange(-180, 180, 0.05)

    lat_idx = np.searchsorted(lat_grid, tile_lat)
    lon_idx = np.searchsorted(lon_grid, tile_lon)

    # Get 2x2 grid of SST values for the date
```

```
    date_str = tile_datetime.strftime('%Y-%m-%d')
    sst_grid = load_coraltemp(date_str)  # From pre-downloaded NetCDF

    # Bilinear interpolation
    sst_value = bilinear_interp(
        sst_grid[lat_idx-1:lat_idx+1, lon_idx-1:lon_idx+1],
        tile_lat, tile_lon,
        lat_grid[lat_idx-1:lat_idx+1],
        lon_grid[lon_idx-1:lon_idx+1]
    )
    return sst_value
```

2. **Tide Height Alignment (CHS):**

```
def get_tide(tile_lat, tile_lon, tile_datetime):
    # Find nearest tide prediction station
    stations = load_chs_stations()  # ~400 stations in BC
    distances = haversine(tile_lat, tile_lon,
                          stations['lat'], stations['lon'])
    nearest_station = stations[np.argmin(distances)]

    # Load tide predictions for that station
    tide_data = load_tide_predictions(nearest_station['id'],
                                      tile_datetime.date())

    # Interpolate to exact image acquisition time
    # Tide predictions are at 6-minute intervals
    tide_height = np.interp(
        tile_datetime.timestamp(),
        tide_data['timestamps'],
        tide_data['heights']
    )
    return tide_height
```

3. **Ocean Color Alignment (Sentinel-3 OLCI):**

```
def get_ocean_color(tile_lat, tile_lon, tile_datetime, product='CHL'):
    # Sentinel-3 OLCI is 300m resolution, ~daily but with gaps due to clouds
    # Search for nearest cloud-free observation within +/- 3 days
    for day_offset in [0, -1, 1, -2, 2, -3, 3]:
        search_date = tile_datetime + timedelta(days=day_offset)
        olci_data = load_olci(search_date, product)

        if olci_data is not None:
            # Check if tile location has valid (non-cloud) data
            value = bilinear_interp(olci_data, tile_lat, tile_lon)
            if not np.isnan(value):
                return value, day_offset  # Return value and temporal offset
```

```
    # No valid data found within window
    return np.nan, None
```

2.3.5 Handling Missing Context Data

Why Data Gaps Occur:

Data Source	Gap Cause	Typical Frequency
SST (CoralTemp)	Rare (composite product)	<1% of days
Tide predictions	Station coverage gaps	5–10% of remote locations
Sentinel-3 OLCI	Clouds, sun glint, swath gaps	30–50% of coastal observations
Bathymetry	Survey coverage gaps	10% of remote areas

Missing Data Strategies (Hierarchical Fallback):
For each feature with potential gaps, we implement a cascade of fallback strategies:
Strategy 1: Temporal Interpolation (for time-varying features)
If data missing at exact time, search nearby times:

```
def get_with_temporal_fallback(lat, lon, datetime, max_days=7):
    # Try exact date first
    value = get_value(lat, lon, datetime)
    if not np.isnan(value):
        return value, 'exact'

    # Try expanding window
    for offset in range(1, max_days + 1):
        for direction in [-1, 1]:
            search_dt = datetime + timedelta(days=offset * direction)
            value = get_value(lat, lon, search_dt)
            if not np.isnan(value):
                return value, f'offset_{offset * direction}d'

    return np.nan, 'missing'
```

Strategy 2: Spatial Interpolation (for spatially sparse data)
If nearest station/grid point has no data, interpolate from surrounding points:

```
def get_with_spatial_fallback(lat, lon, datetime, product):
    # Try direct lookup
    value = get_value(lat, lon, datetime)
    if not np.isnan(value):
        return value, 'direct'

    # Find k nearest valid observations and inverse-distance weight
    neighbors = find_k_nearest_valid(lat, lon, datetime, k=4)
    if len(neighbors) >= 2:
        weights = 1.0 / neighbors['distances']
        value = np.average(neighbors['values'], weights=weights)
```

```
        return value, 'idw_interpolated'

    return np.nan, 'missing'
```

Strategy 3: Climatological Fallback (for persistent gaps)
Use long-term average for that location and time of year:

```
def get_with_climatology_fallback(lat, lon, datetime, product):
    value, method = get_with_temporal_fallback(lat, lon, datetime)
    if not np.isnan(value):
        return value, method

    # Use climatological mean for this location and day-of-year
    doy = datetime.timetuple().tm_yday
    climatology = load_climatology(product)  # Pre-computed from 8-year archive
    clim_value = bilinear_interp(climatology[doy], lat, lon)

    if not np.isnan(clim_value):
        return clim_value, 'climatology'

    return np.nan, 'missing'
```

Strategy 4: Imputation with Flag (last resort)
If all else fails, impute with a default value but flag the feature:

```
def get_with_imputation(lat, lon, datetime, product):
    value, method = get_with_climatology_fallback(lat, lon, datetime)
    if not np.isnan(value):
        return value, method, False  # False = not imputed

    # Impute with global mean from training set
    imputed_value = GLOBAL_MEANS[product]
    return imputed_value, 'imputed', True  # True = was imputed
```

Feature-Specific Fallback Configuration:

Feature	Primary	Fallback 1	Fallback 2	Last Resort
SST	Exact date	±3 day window	Monthly climatology	Training mean (14.2°C)
Tide Height	Nearest station	2nd nearest	Interpolate 3 stations	0.0m (mean tide)
Chlorophyll-a	Exact date	±7 day window	Seasonal climatology	Training mean
Turbidity (Kd490)	Exact date	±7 day window	Seasonal climatology	Training mean
Bathymetry	Direct lookup	IDW from neighbors	Regional mean depth	-20m (typical kelp depth)

2.3.6 Tracking Data Quality in Context Vectors

Augmented Context Vector with Quality Flags:

To inform the k-NN matching about data quality, we augment the context vector with quality indicators:

$$\mathbf{c}_t^{\text{aug}} = [\mathbf{c}_t, \underbrace{q_{\text{SST}}, q_{\text{tide}}, q_{\text{chl}}, q_{\text{turb}}, q_{\text{bathy}}}_{\text{5 quality flags}}]$$

where $q_i \in \{0, 1, 2, 3\}$ indicates:

- 0 = Exact match (highest quality)
- 1 = Temporal/spatial interpolation (good quality)
- 2 = Climatology fallback (moderate quality)
- 3 = Imputed (low quality)

Why this matters: When querying the memory bank, we want to match tiles that have similar *quality* of context data. A prediction made with imputed SST should be compared to other predictions with imputed SST, not to predictions with exact SST measurements.

2.3.7 Expected Data Availability Analysis

Based on preliminary analysis of BC coastal zone (2017–2024):

Feature	Exact Match	Interpolated	Climatology/Imputed
SST	95%	4%	1%
Tide Height	85%	12%	3%
Chlorophyll-a	45%	35%	20%
Turbidity (Kd490)	45%	35%	20%
Bathymetry	90%	8%	2%

Key Insight: Ocean color products (Chl, Kd490) have lowest availability due to coastal cloud cover. This is acceptable because:

1. These features are supplementary (spectral features from imagery are primary)
2. Climatology provides reasonable seasonal baseline
3. Quality flags ensure k-NN matches consider data quality

2.3.8 Validation of Context Feature Alignment

Sanity Checks to Perform:

1. **SST Range Check**: BC coastal SST should be 5–20°C. Flag values outside this range.
2. **Tide Consistency**: Compare predicted tide to satellite-observed water line position (where visible).
3. **Spatial Coherence**: Adjacent tiles should have similar SST, Chl values (unless at front/boundary).

4. **Seasonal Patterns**: DOY-stratified distributions should show expected seasonal cycles.

Diagnostic Visualizations to Generate:

- Maps of context feature coverage (what percentage of tiles have exact vs. interpolated data)
- Time series of data availability by feature
- Scatter plots of context features vs. tile location (check for spatial biases)
- Histograms of feature values by quality flag (check if imputed values are reasonable)

2.3.9 How Context Features Are Used in the Framework

Step 1: Storage in Memory Bank

When a prediction receives a label (reward), we store:

$$\mathcal{M} \leftarrow \mathcal{M} \cup \{(\mathbf{c}_t, \mathbf{p}_{\text{final}}, R)\}$$

This records: "Under context $\mathbf{c}_t$, using probability vector $\mathbf{p}_{\text{final}}$, we achieved reward R"

Step 2: Retrieval for New Tiles

When a new tile arrives with context $\mathbf{c}_{\text{new}}$:

1. Query memory bank for k nearest neighbors by L2 distance:

$$\mathcal{N}_k(\mathbf{c}_{\text{new}}) = \text{arg-}k\text{-min}_j \|\mathbf{c}_{\text{new}} - \mathbf{c}_j\|_2$$

2. Compute distance-based weights:

$$w_j = \exp\left(-\frac{\|\mathbf{c}_{\text{new}} - \mathbf{c}_j\|_2^2}{2\sigma^2}\right)$$

   where σ is a bandwidth hyperparameter

3. Compute local probability estimate (reward-weighted average):

$$\mathbf{p}_{\text{local}} = \frac{\sum_{j \in \mathcal{N}_k} w_j \cdot R_j \cdot \mathbf{p}_j}{\sum_{j \in \mathcal{N}_k} w_j \cdot R_j}$$

4. Blend with global policy:

$$\mathbf{p}_{\text{final}} = \alpha \cdot \mathbf{p}_{\text{global}} + (1 - \alpha) \cdot \mathbf{p}_{\text{local}}$$

Intuition: If the new tile has high turbidity (low blue/green ratio, high Kd490), and the memory bank contains entries from similar turbid conditions where Random Forest performed well, then $\mathbf{p}_{\text{local}}$ will favor Random Forest.

2.3.10 Concrete Example: Context-Driven Heuristic Selection

Scenario: Two tiles from the same location, different conditions

Tile A (Clear Summer Day):

- SST = 14°C, Turbidity (Kd490) = 0.05, Tide = +1.5m (high)
- Context vector $\mathbf{c}_A$ is similar to memory entries where NDVI threshold worked well
- $\mathbf{p}_{\text{local}}$ = [0.6, 0.2, 0.05, 0.05, 0.02, 0.02, 0.02, 0.01, 0.01, 0.01, 0.005, 0.005]
- Heuristic 1 (NDVI) likely selected

Tile B (Turbid Winter Day):

- SST = 8°C, Turbidity (Kd490) = 0.25, Tide = -0.5m (low)
- Context vector $\mathbf{c}_B$ is similar to memory entries where Random Forest worked well
- $\mathbf{p}_{\text{local}}$ = [0.05, 0.05, 0.02, 0.02, 0.5, 0.2, 0.1, 0.02, 0.01, 0.01, 0.01, 0.01]
- Heuristic 5 (Random Forest) likely selected

2.4 Data Stream 3: Ground Truth Labels

Purpose: Labels provide the **reward signal** that drives LA learning. Without labels, the LA cannot update its policy.

Critical Characteristics:

- **Sparse**: Only 1–10% of tiles will ever receive labels
- **Delayed**: Labels arrive days to months after the satellite image was acquired
- **Heterogeneous**: Labels come from different sources with different accuracies

2.4.1 Label Sources - Detailed Breakdown

Source 1: ShoreZone Oblique Photography

- **What it is**: Aerial photography of BC coastline taken from helicopter/aircraft
- **Coverage**: Entire BC coastline, but updated only every 10–15 years
- **Format**: Classified polygons indicating kelp presence/absence
- **Accuracy**: High (expert interpretation), but temporal mismatch with satellite
- **Access**: BC Marine Conservation Atlas, Data BC
- **Use in framework**: Historical baseline, not for real-time learning

Source 2: WorldView High-Resolution Classifications

- **What it is**: Commercial satellite imagery (1.2–2.6m resolution) with expert kelp classification
- **Coverage**: Limited areas (e.g., Cowichan Bay, Barkley Sound)

- **Format**: Rasterized kelp masks, can be resampled to Sentinel-2 resolution
- **Accuracy**: 86.9% overall accuracy (Schroeder 2020)
- **Temporal alignment**: Can match within ±5 days of Sentinel-2 acquisition
- **Use in framework**: High-quality labels for specific regions

Source 3: In-Situ Surveys (SCUBA, Shoreline)

- **What it is**: Direct observation by divers or shore-based surveyors
- **Coverage**: Point observations at specific coordinates
- **Format**: GPS coordinates with kelp presence/absence/density
- **Accuracy**: Very high (ground truth), but point-based, not pixel-based
- **Challenge**: Must convert point observations to tile-level labels
- **Use in framework**: Validation of heuristic accuracy at surveyed locations

Source 4: Citizen Science (iNaturalist, Kelp Projects)

- **What it is**: Geotagged observations from public contributors
- **Coverage**: Opportunistic, biased toward accessible areas
- **Format**: GPS coordinates with photos, sometimes species ID
- **Accuracy**: Variable (requires quality filtering)
- **Use in framework**: Supplementary labels, especially for accessible coastline

Source 5: UAV/Drone Surveys

- **What it is**: Low-altitude aerial imagery from unoccupied aerial vehicles
- **Coverage**: Very limited spatial extent, but very high resolution (cm-scale)
- **Format**: Orthomosaics with kelp classification
- **Accuracy**: 93% (Cavanaugh et al. 2021)
- **Use in framework**: High-quality local validation, especially for accuracy assessment

2.4.2 Label Processing Pipeline

Step 1: Temporal Matching

For each label source, find corresponding Sentinel-2 tiles:

- Match within temporal window $\pm\tau$ days (e.g., $\tau = 7$)
- Rationale: Kelp canopy can change rapidly; too large τ introduces noise
- For ShoreZone (annual surveys), use larger τ for summer period only

Step 2: Spatial Resampling
Convert label source to Sentinel-2 tile grid:

- WorldView: Downsample from 2.6m to 10m using majority vote within each 10m pixel
- Point observations: Buffer to 30m radius, assign to overlapping tiles
- Polygons: Rasterize to 10m grid

Step 3: Quality Filtering
Remove low-quality labels:

- Citizen science: Require photo verification, minimum confidence score
- Old surveys: Weight by age (temporal decay)
- Cloud-affected: Reject if Sentinel-2 tile had >10% cloud at label time

Step 4: Label Format Standardization
Final label format for each tile:

$$(\ell, t, \mathbf{y}) \quad \text{where } \mathbf{y} \in \{0, 1, \text{unknown}\}^{256\times256}$$

- ℓ: Tile location (lat/lon of center)
- t: Timestamp of the satellite image being labeled
- $\mathbf{y}$: Binary mask (0 = no kelp, 1 = kelp, unknown = no label for this pixel)

2.4.3 How Labels Enter the Framework

The Asynchronous Label Arrival Process:

1. **Day 0**: Sentinel-2 tile $\mathbf{x}_t$ acquired at location ℓ
2. **Day 0**: Framework processes tile, selects heuristic h_i, produces prediction $\hat{\mathbf{y}}_t$
3. **Day 0**: Prediction stored in buffer: $\mathcal{B} \leftarrow \mathcal{B} \cup \{(t, \ell, \mathbf{c}_t, i, \hat{\mathbf{y}}_t)\}$
4. **Day** δ: Label $\mathbf{y}_t$ arrives (from survey, WorldView, etc.)
5. **Day** δ: System looks up buffered prediction for (ℓ, t)
6. **Day** δ: Compute reward $R = \text{IoU}(\hat{\mathbf{y}}_t, \mathbf{y}_t)$
7. **Day** δ: Apply temporal decay $R_{\text{eff}} = R \cdot \gamma^{\delta}$
8. **Day** δ: Update LA policy (Component 1) and memory bank (Component 2)
9. **Day** δ: Remove entry from buffer

Why δ varies:

- WorldView tasking: 1–7 days (commercial satellite can be tasked quickly)
- Citizen science: 1–30 days (depends on when someone visits the area)
- Planned surveys: 30–180 days (surveys scheduled months in advance)
- ShoreZone: Years (archival data)

3 Sparse Label Handling - Complete Explanation

This section provides an exhaustive explanation of Component 3 (Sparse Label Handling), which is essential for the framework to learn from operational monitoring conditions.

3.1 The Fundamental Problem

Standard Machine Learning Assumption:
In typical supervised learning:

- Training data: $\{(\mathbf{x}_1, \mathbf{y}_1), (\mathbf{x}_2, \mathbf{y}_2), \ldots, (\mathbf{x}_N, \mathbf{y}_N)\}$
- Every input $\mathbf{x}_i$ has a corresponding label $\mathbf{y}_i$
- Labels are available *before* model is deployed

Operational Kelp Monitoring Reality:

- We acquire 100,000+ satellite tiles per year
- Ground truth surveys can validate maybe 1,000–5,000 of these
- That's 1–5% label availability
- Labels arrive *after* the system has already made predictions
- Delay ranges from days (rapid UAV survey) to months (planned research cruise)

The Challenge:
How can a system learn and improve when:

1. 90–99% of predictions never receive feedback?
2. Feedback for the remaining 1–10% arrives unpredictably?
3. By the time feedback arrives, the system has processed many more tiles?

3.2 The Sparse Label Handling Solution

Core Idea: Maintain a buffer of predictions "awaiting validation." When a label eventually arrives, look up the corresponding prediction and compute the reward.

This is analogous to:

- *Experience replay* in deep reinforcement learning (but for sparse rewards)
- *Credit assignment* in delayed reward settings
- *Asynchronous SGD* in distributed training (but for label arrival)

3.2.1 Data Structures

Structure 1: Prediction Buffer $\mathcal{B}$

A FIFO (First-In-First-Out) queue storing predictions awaiting validation:

$$\mathcal{B} = \{(t_1, \ell_1, \mathbf{c}_1, i_1, \hat{\mathbf{y}}_1), (t_2, \ell_2, \mathbf{c}_2, i_2, \hat{\mathbf{y}}_2), \ldots\}$$

Each entry contains:

Field	Type	Description
t	timestamp	When the satellite image was acquired
ℓ	(lat, lon)	Location (tile center coordinates)
$\mathbf{c}$	$\mathbb{R}^{29}$	Context feature vector at prediction time
i	int $\in \{1, \ldots, 12\}$	Index of heuristic that was selected
$\hat{\mathbf{y}}$	$\{0,1\}^{256\times256}$	The prediction that was made

Storage requirements:

- Context vector: 29×4 bytes = 116 bytes
- Prediction mask: $256 \times 256 \times 1$ bit = 8 KB (or compressed)
- Metadata: ∼50 bytes
- Per entry: ∼8 KB
- For 100,000 buffered predictions: ∼800 MB

Structure 2: Hash Map $\mathcal{L}$

For O(1) lookup when labels arrive:

$$\mathcal{L} : (\ell, t) \mapsto \text{buffer_index}$$

- Key: (location, timestamp) tuple
- Value: Index into buffer $\mathcal{B}$
- Implementation: Python dict or Redis for distributed systems

3.2.2 Detailed Algorithm

Algorithm 1: Prediction and Buffering (runs for every tile)

```
def process_tile(x_t, t, location):
    # Step 1: Extract context features
    c_t = extract_context(x_t, t, location)

    # Step 2: Get selection probability (Component 2)
    p_local = query_memory_bank(c_t, k=10)
    alpha = compute_blending_coefficient(c_t)
    p_final = alpha * p_global + (1 - alpha) * p_local
```

```
    # Step 3: Select heuristic
    i = sample_categorical(p_final)

    # Step 4: Apply heuristic to get prediction
    y_hat = heuristics[i].predict(x_t)

    # Step 5: Store in buffer for future validation
    buffer_entry = (t, location, c_t, i, y_hat)
    buffer.append(buffer_entry)
    lookup_map[(location, t)] = len(buffer) - 1

    # Step 6: Clean old entries (beyond T_max)
    clean_old_entries(T_max=180)  # 6 months

    return y_hat
```

Algorithm 2: Label Processing (runs when label arrives)

```
def process_label(y_true, t_label, location):
    # Step 1: Look up the original prediction
    if (location, t_label) not in lookup_map:
        # No matching prediction (expired or never made)
        return

    buffer_index = lookup_map[(location, t_label)]
    t_pred, loc, c_t, i, y_hat = buffer[buffer_index]

    # Step 2: Compute raw reward (IoU)
    intersection = np.sum(y_hat & y_true)
    union = np.sum(y_hat | y_true)
    R = intersection / union if union > 0 else 0.0

    # Step 3: Compute delay and apply temporal decay
    delta = (datetime.now() - t_pred).days
    R_effective = R * (gamma ** delta)  # gamma = 0.99

    # Step 4: Update global policy (Component 1)
    # L_{R-I} update for action i with reward R_effective
    p_global[i] += lambda_lr * R_effective * (1 - p_global[i])
    for j in range(12):
        if j != i:
            p_global[j] -= lambda_lr * R_effective * p_global[j]

    # Step 5: Update memory bank (Component 2)
    memory_bank.add(c_t, p_final_at_prediction_time, R)

    # Step 6: Remove from buffer
    del buffer[buffer_index]
    del lookup_map[(location, t_label)]
```

3.2.3 Why Temporal Decay?

The Problem with Stale Feedback:

Consider a prediction made on January 1st for a tile showing kelp. A label arrives on July 1st (180 days later). By July:

- Kelp may have grown or died (bull kelp is annual)
- Environmental conditions are completely different
- The heuristic that worked in January may not be optimal in July

If we treat this feedback equally with recent feedback, we contaminate the learning signal.

The Temporal Decay Solution:

Weight rewards by recency:

$$R_{\text{effective}} = R \cdot \gamma^{\delta}$$

where δ is delay in days and $\gamma \in (0.95, 0.999)$.

Example with $\gamma = 0.99$:

Delay δ	**Decay Factor γ^{δ}**	**Interpretation**
1 day	0.99	Almost full weight
7 days	0.93	93% weight
30 days	0.74	74% weight
90 days	0.41	41% weight
180 days	0.16	16% weight

Intuition: Recent feedback gets nearly full weight; old feedback gets downweighted but not completely ignored.

3.2.4 Buffer Management

Maximum Buffer Age T_{max}:

Entries older than T_{max} (default: 180 days / 6 months) are removed without update:

```
def clean_old_entries(T_max):
    current_time = datetime.now()
    to_remove = []
    for idx, (t, loc, c, i, y_hat) in enumerate(buffer):
        if (current_time - t).days > T_max:
            to_remove.append((loc, t))

    for (loc, t) in to_remove:
        del lookup_map[(loc, t)]

    buffer = [e for e in buffer if (current_time - e[0]).days <= T_max]
```

Rationale:

- After 6 months, conditions have changed enough that feedback is no longer informative
- Prevents unbounded buffer growth

- Aligns with kelp phenology (annual cycle)

Memory Efficiency:
For operational deployment:

- Don't store full prediction mask; store compressed version or hash
- Use database (PostgreSQL, Redis) instead of in-memory for large-scale
- Index by (location, time) for efficient queries

3.2.5 Handling Edge Cases

Case 1: Label arrives for non-existent prediction
Can happen if:

- Prediction was made before system started (historical labels)
- Prediction was cleaned from buffer (too old)
- Location/time mismatch due to coordinate precision

Solution: Skip update, log for monitoring.
Case 2: Multiple labels for same tile
Can happen if multiple validation sources cover the same area.
Solution: Use highest-quality label (based on source reliability ranking).
Case 3: Partial labels (not all pixels labeled)
Common with point observations and small UAV footprints.
Solution: Compute IoU only over labeled pixels:

$$R = \frac{|\hat{\mathbf{y}}_{\text{labeled}} \cap \mathbf{y}_{\text{labeled}}|}{|\hat{\mathbf{y}}_{\text{labeled}} \cup \mathbf{y}_{\text{labeled}}|}$$

Case 4: No labels ever arrive (99% of tiles)
This is the normal case! The $L_{\text{R-I}}$ "inaction" property handles it:

- No label $\Rightarrow$ No update
- Probabilities remain unchanged
- System continues using best-known policy

This is the key advantage of $L_{\text{R-I}}$ over methods that require constant feedback.

3.2.6 Experimental Validation of Sparse Label Handling

Protocol C experiments will test:

1. **Sparsity levels**: What happens as label availability decreases?
    - 100% labels (baseline)
    - 50% labels (moderately sparse)
    - 10% labels (sparse)

- 1% labels (very sparse)

2. **Delay distributions**: How does delay affect learning?
   - Immediate (0 days) - baseline
   - Short delay: $\delta \sim U[1, 7]$ days
   - Medium delay: $\delta \sim U[7, 30]$ days
   - Long delay: $\delta \sim U[30, 180]$ days
   - Mixed: Realistic distribution from actual validation timelines
3. **Metrics to report**:
   - **Convergence rate**: How many labels until 90% of asymptotic performance?
   - **Final performance**: IoU/F1 after all labels processed
   - **Stability**: Variance in heuristic selection over time
   - **Comparison**: LA vs. baseline that discards delayed labels vs. baseline that ignores delay

4 Methodology Flow - Complete Walkthrough

Step-by-Step Processing of One Tile:

Step 1: Input Acquisition

- Sentinel-2 tile $\mathbf{x}_t$ arrives ($256 \times 256 \times 5$ bands)
- Timestamp t, location ℓ
- Already preprocessed: cloud-masked, land-masked

Step 2: Context Feature Extraction

- Compute spectral statistics: mean, variance per band (10 features)
- Compute spatial texture: entropy, homogeneity (2 features)
- Look up temporal context: day of year, days since last observation (2 features)
- Query environmental data: SST from NOAA, tide from CHS (2 features)
- Extract geographic: lat, lon, distance to shore (3 features)
- Result: context vector $\mathbf{c}_t \in \mathbb{R}^D$ where $D \approx 19$

Step 3: k-NN Retrieval (Component 2)

- Query memory bank $\mathcal{M}$ for k nearest neighbors to $\mathbf{c}_t$
- Use FAISS for $O(\log M)$ approximate search
- Retrieve: $\{(\mathbf{c}_j, \mathbf{p}_j, R_j)\}$ for $j \in \mathcal{N}_k(\mathbf{c}_t)$

Step 4: Probability Blending

- Compute local probability: $\mathbf{p}_{\text{local}} = \frac{\sum w_j R_j \mathbf{p}_j}{\sum w_j R_j}$
- Compute blending coefficient $\alpha(\mathbf{c}_t)$ based on neighbor confidence
- Blend: $\mathbf{p}_{\text{final}} = \alpha \cdot \mathbf{p}_{\text{global}} + (1 - \alpha) \cdot \mathbf{p}_{\text{local}}$

Step 5: Heuristic Selection

- Sample heuristic index $i \sim \text{Categorical}(\mathbf{p}_{\text{final}})$
- Select heuristic h_i from pool $\mathcal{H}$

Step 6: Prediction

- Apply heuristic: $\hat{\mathbf{y}}_t = h_i(\mathbf{x}_t)$
- Result: binary mask (256×256) indicating kelp/non-kelp

Step 7: Buffering (Component 3)

- Add to prediction buffer: $\mathcal{B} \leftarrow \mathcal{B} \cup \{(t, \mathbf{x}_t, \mathbf{c}_t, i, \hat{\mathbf{y}}_t)\}$
- Update hash map: $\mathcal{L}[(\ell, t)] \leftarrow$ buffer_index

Step 8: Label Arrival (Asynchronous)

- Ground truth $\mathbf{y}_t$ arrives for location ℓ, time t
- Look up: buffer_index $= \mathcal{L}[(\ell, t)]$
- Retrieve: $(t, \mathbf{x}_t, \mathbf{c}_t, i, \hat{\mathbf{y}}_t)$

Step 9: Reward Computation

- Compute IoU: $R = \frac{|\hat{\mathbf{y}}_t \cap \mathbf{y}_t|}{|\hat{\mathbf{y}}_t \cup \mathbf{y}_t|}$
- Compute delay: $\delta = t_{\text{now}} - t$
- Apply temporal decay: $R_{\text{eff}} = R \cdot \gamma^{\delta}$

Step 10: Policy Update (Component 1)

- Update global policy using $\text{L}_{\text{R-I}}$ with reward R_{eff} for action i:

$$p_i \leftarrow p_i + \lambda \cdot R_{\text{eff}} \cdot (1 - p_i) \tag{5}$$

$$p_j \leftarrow p_j - \lambda \cdot R_{\text{eff}} \cdot p_j, \quad \forall j \neq i \tag{6}$$

Step 11: Memory Bank Update (Component 2)

- Add to memory: $\mathcal{M} \leftarrow \mathcal{M} \cup \{(\mathbf{c}_t, \mathbf{p}_{\text{final}}, R)\}$
- If $|\mathcal{M}| > M_{\text{max}}$, remove oldest entry (FIFO)

Step 12: Cleanup

- Remove from buffer: $\mathcal{B} \leftarrow \mathcal{B} \setminus \{(t, \ldots)\}$
- Remove from hash map: $\mathcal{L}[(\ell, t)] \leftarrow$ null

5 Comprehensive Q&A

5.1 Questions About Learning Automata

Q1: Why Learning Automata instead of standard Reinforcement Learning (Q-learning, Policy Gradient)?

A: Several reasons make LA more suitable for this problem:

1. **Sparse feedback handling**: LA (especially $L_{R\text{-}I}$) has the "inaction" property—when no feedback is received, probabilities remain unchanged. Standard RL methods require feedback at every timestep or use bootstrapping (temporal difference), which can introduce instability with sparse feedback.

2. **Convergence guarantees**: $L_{R\text{-}I}$ is proven ϵ-optimal in all stationary environments. Q-learning converges under stricter conditions (all state-action pairs visited infinitely often).

3. **Simplicity**: LA maintains a probability vector of size r (12 heuristics). Q-learning would need a Q-table or neural network mapping (state, action) $\rightarrow$ value, which is more complex.

4. **Domain fit**: The problem naturally fits LA formulation—we're selecting among discrete actions (heuristics) based on sparse rewards (validation labels).

5. **My prior expertise**: I have published work on Learning Automata, specifically Object Migration Automata (OMA). This gives me theoretical grounding.

However, I acknowledge that deep RL could be explored as an extension, especially for learning the context features or heuristic parameters.

Q2: What if the optimal heuristic changes over time (non-stationarity)?

A: This is a key consideration, and the framework handles it through:

1. **Context-aware adaptation (Component 2)**: The memory bank allows different heuristic preferences for different contexts. Even if optimal heuristic changes globally, context-specific adaptation can track local optimality.

2. **Temporal decay (Component 3)**: Old feedback is downweighted by γ^{δ}, reducing influence of stale information. This provides implicit forgetting.

3. **MSE/PSE theory**: LA theory includes analysis for Markovian Switching Environments (MSE) and Periodic Switching Environments (PSE). My prior work showed PEOMA outperforms EOMA by order of magnitude in MSEs. I can incorporate these techniques.

4. **Potential extensions**:
   - Sliding window for global policy (only recent rewards count)
   - Explicit change detection with policy reset
   - Pursuit algorithms that track running estimates

Q3: How do you choose the learning rate λ?

A: This is a hyperparameter to be tuned. Key considerations:

1. **Theory**: ϵ-optimality holds as $\lambda \rightarrow 0$, but convergence becomes infinitely slow. Practical values: $\lambda \in [0.01, 0.2]$.

2. **Trade-off**: Small λ = more stable but slower adaptation. Large λ = faster adaptation but more noise sensitivity.

3. **Plan**: Grid search over $\lambda \in \{0.01, 0.05, 0.1, 0.2\}$ in validation experiments.

4. **Adaptive rate**: Could decay λ over time (start large for exploration, decrease for stability).

Q4: Can you explain the difference between Fixed Structure (FSSA) and Variable Structure (VSSA) automata?
A:
FSSA (e.g., Tsetlin automaton):

- Maintains N memory states per action
- State transitions are deterministic given current state and feedback
- Reward: move deeper into action's memory
- Penalty: move toward boundary where action changes
- Action selection is deterministic from state

VSSA (e.g., $\mathbf{L_{R\text{-}I}}$):

- Directly maintains probability vector $\mathbf{p}$
- Updates probabilities using continuous equations
- More analytically tractable
- Basis for modern LA applications

I use VSSA (specifically $L_{R\text{-}I}$) because it's more flexible for continuous S-model rewards and easier to analyze mathematically.

5.2 Questions About the Framework

Q5: Why a 3-tier heuristic pool? Why not just use the best deep learning model?
A:

1. **Complementary failure modes**: Deep learning models can overfit to training distribution. Spectral indices are robust to distribution shift but less expressive. Having both allows the LA to select the right tool for each situation.

2. **Computational efficiency**: Spectral indices are 100-1000× faster than deep learning. For clear conditions where indices suffice, we save computation. Carbon footprint matters.

3. **Interpretability**: Indices are interpretable ("kelp has NDVI > 0.1"). This is valuable for validation with domain experts and TEK holders.

4. **Empirical evidence**: Yong et al. (2005) showed 85% accuracy in predicting optimal segmentation algorithm based on image features. Different algorithms excel under different conditions.

5. **Ablation planned**: I will explicitly test removing each tier to quantify contribution.

Q6: How does the k-NN retrieval scale with memory bank size?
A:

1. **FAISS**: I use Facebook AI Similarity Search (FAISS) for efficient approximate nearest neighbor. With IVF (Inverted File) indexing, query time is $O(\log M)$ where M is memory bank size.

2. **Memory management**: Memory bank has maximum size $M_{\max}$. When exceeded, oldest entries are removed (FIFO). This bounds memory usage and ensures recency.

3. **Expected scale**: For BC coastline, expect ∼100,000 tiles over 8 years. With 10% labeled, ∼10,000 memory entries. FAISS handles millions of entries easily.

4. **Alternative**: Could use locality-sensitive hashing (LSH) or clustering-based indexing if scale becomes issue.

Q7: What if there are no similar contexts in the memory bank (cold start)?
A:

1. **Adaptive blending**: When few neighbors found or distances are large, $\alpha(\mathbf{c}) \to 1$, favoring global policy. The global policy provides reasonable default.

2. **Initialization**: Start with uniform global policy $p_i = 1/12$. All heuristics get tried with equal probability initially.

3. **Bootstrapping**: Can pre-populate memory bank with historical data if available (e.g., prior studies, WorldView classifications).

4. **Exploration-exploitation**: Early on, more exploration (uniform selection). As memory grows, more exploitation (local adaptation).

Q8: How do you handle the case where multiple heuristics perform similarly well?
A:

1. **$\mathbf{L_{R\text{-}I}}$ behavior**: When multiple heuristics receive similar rewards, probability mass distributes among them. The automaton doesn't necessarily converge to a single action—it can maintain a mixture.

2. **Practical benefit**: If NDVI and FAI both work well for clear water, the system might use either. This provides robustness.

3. **Multi-objective extension**: Can add computational cost as secondary objective. Between equally accurate heuristics, prefer faster/cheaper one.

4. **Ensemble possibility**: Could extend framework to select multiple heuristics and ensemble their predictions. Not in current design but possible extension.

Q9: How sensitive is the framework to the choice of context features?
A:

1. **Ablation planned**: Will systematically test subsets of context features to identify most informative.

2. **Prior knowledge**: Based on literature, expect SST, turbidity, and tide to be most important (Schroeder 2020, Timmer 2024).

3. **Feature learning**: Could learn context representation via autoencoder or contrastive learning. This is a potential extension.

4. **Distance metric**: Currently using L2 distance. Could learn metric via Mahalanobis distance or neural metric learning.

5.3 Questions About Experimental Design

Q10: How large is the BC Bull Kelp dataset?
A:

1. **Imagery**: Sentinel-2 covers BC coast. At 10m resolution, 256×256 tiles cover 2.56 km^2. For 26,000 km coastline with 1 km buffer, approximately 10,000 km^2 coverage. That's ∼4,000 tiles per acquisition.

2. **Temporal**: 5-day revisit × 8 years = ∼580 acquisitions. Not all are usable (clouds, winter). Estimate 30% usable = ∼175 acquisitions.

3. **Total**: ∼700,000 tile-observations. After cloud filtering, expect 100,000–200,000 usable tiles.

4. **Labels**: Ground truth is sparse. Estimate 1–10% labeled = 1,000–20,000 labeled tiles. This is a key challenge the framework addresses.

Q11: How do you ensure fair comparison with baselines?
A:

1. **Same data splits**: All methods see exactly the same train/val/test splits. Use Protocol A (temporal), Protocol B (spatial).

2. **Same computational budget**: For label efficiency curves, compare at same number of labels, not same training time.

3. **Hyperparameter tuning**: All methods get same hyperparameter search budget on validation set.

4. **Multiple seeds**: Report mean ± std over 5+ random seeds.

5. **Statistical tests**: Use paired t-test or Wilcoxon signed-rank for significance.

Q12: What are your baselines?
A:

1. **Fixed heuristic**: Each of the 12 heuristics used alone (no adaptation)

2. **Best single heuristic**: Oracle that always selects the globally best heuristic (upper bound for fixed approach)

3. **Random selection**: Uniform random selection among heuristics

4. **Supervised CNN**: U-Net trained on all available labels with standard training

5. **Online learning**: Baseline that updates randomly when labels missing
6. **Oracle**: Per-tile best heuristic (upper bound for any selection method)
7. **Majority vote ensemble**: All heuristics applied, majority vote

Q13: How do you simulate sparse and delayed labels?
A:

1. **Sparsity**: Randomly withhold labels. Test 10%, 50%, 90% withheld.
2. **Delay distributions**:
   - Uniform: $\delta \sim U[1, 180]$ days
   - Exponential: $\delta \sim \text{Exp}(\mu = 30)$ days
   - Realistic: Based on actual validation timeline if available
3. **Biased sampling**: Test if label availability correlates with conditions (e.g., more labels for clear conditions). This tests robustness to selection bias.
4. **Metrics**: Convergence rate, final performance, stability over time.

5.4 Questions About Practical Considerations

Q14: What about computational cost? Can this run operationally?
A:

1. **Inference cost**: Depends on selected heuristic. Indices: milliseconds per tile. Deep learning: seconds per tile on GPU.
2. **LA overhead**: Negligible. k-NN lookup is $O(\log M)$. Probability update is $O(r) = O(12)$.
3. **Memory**: Global policy: 12 floats. Memory bank: $M \times (D+12+1)$ floats. For $M = 10,000$, $D = 19$: $\sim$1.3 MB.
4. **Operational feasibility**: Yes. Main cost is running heuristics, not the LA. Could run on standard server, potentially edge device for simple heuristics.
5. **Multi-objective**: R_{speed} reward encourages selection of faster heuristics when accuracy is similar.

Q15: How does this compare to transfer learning or domain adaptation?
A:

1. **Complementary approaches**: Transfer learning adapts model parameters. LA adapts model *selection*. Could combine: pre-train heuristics via transfer learning, then use LA for selection.
2. **Different assumptions**: Transfer learning assumes source and target domains are related. LA makes no such assumption—it learns from feedback in target domain directly.
3. **Label requirements**: Fine-tuning for transfer needs labeled target data. LA can start with zero target labels and learn from sparse feedback.

4. **Potential extension**: Domain-adaptive heuristics in the pool (e.g., DANN-trained U-Net). LA selects among adapted and non-adapted models.

Q16: What if all heuristics perform poorly in some conditions?
A:

1. **Detection**: Can track uncertainty or confidence. High uncertainty = all heuristics unreliable.
2. **Flagging**: Output "low confidence" flag for tiles where all heuristics disagree or have historically poor performance.
3. **Feedback to operators**: Alert that certain conditions (e.g., very high turbidity) are beyond system capability.
4. **Heuristic pool expansion**: If systematic failure mode identified, add new heuristic designed for that condition.
5. **Graceful degradation**: Better to know you can't detect kelp than to confidently give wrong answer.

5.5 Questions About TEK Integration

Q17: How will you actually integrate TEK? Do you have community partnerships?
A:

1. **Current status**: RQ4 is exploratory. I have not yet established formal partnerships. This is Phase 4 work (after core framework validated).
2. **Approach**: Four-phase protocol:
   - Phase 1: Relationship building through proper channels
   - Phase 2: Co-design integration points with community input
   - Phase 3: Implementation with ongoing engagement
   - Phase 4: Community-led evaluation
3. **Potential partners**: Coastal First Nations (Heiltsuk, Haida, etc.), Ocean Wise, provincial agencies with existing relationships.
4. **Key principles**: Indigenous data sovereignty, Two-Eyed Seeing, benefit sharing, no extractive data collection.
5. **Backup plan**: If community partnerships don't materialize, RQ4 becomes theoretical framework + recommendations rather than empirical study.

Q18: What specific TEK elements could be integrated?
A:

1. **Site importance weights**: Weight evaluation metrics by cultural significance. Prioritize accuracy at culturally important sites.
2. **Temporal priors**: Traditional seasonal calendars inform expected kelp phenology. Can weight predictions by alignment with traditional expectations.

3. **Validation labels**: Elder observations as additional ground truth. Especially valuable for historical baseline.

4. **Anomaly detection**: TEK holders might identify unusual patterns that automated system misses.

5. **Decision support**: Framework outputs inform community-based monitoring and management decisions.

5.6 Questions About Timeline and Scope

Q19: Is October 2027 realistic? That's less than 2 years.
A:

1. **Foundation work done**: I've completed literature review, have working proposal, understand LA theory deeply from prior publications.

2. **Dataset accessible**: Sentinel-2 data is freely available via Google Earth Engine. Ground truth sources identified.

3. **Modular design**: Three components can be developed incrementally. Can validate core framework (Components 1+2) before full sparse label experiments.

4. **Publications strategy**: Four papers target different aspects:

   - Paper 1 (RQ1): Core LA framework
   - Paper 2 (RQ2): Context-aware adaptation
   - Paper 3 (RQ3): Sparse label learning
   - Paper 4 (RQ4): TEK integration

5. **Risk mitigation**: If RQ4 (TEK) doesn't materialize fully, can still complete thesis with RQ1-3.

Q20: What are the main risks to your timeline?
A:

1. **Data quality issues**: If ground truth labels are noisier than expected, may need additional preprocessing or label quality filtering.

2. **Negative results**: If LA doesn't outperform baselines, need to pivot. But even negative results are publishable if well-analyzed.

3. **Computational resources**: Deep learning training requires GPU time. Have access to Compute Canada, but queue times can delay experiments.

4. **TEK partnerships**: Relationship building takes time. May not achieve formal partnerships within timeline.

5. **Scope creep**: Temptation to add more heuristics, more features, more experiments. Must stay focused on core contributions.

5.7 Challenging/Skeptical Questions

Q21: Isn't this just an ensemble method with extra steps?
A:

1. **Key difference**: Ensembles typically combine all models (e.g., averaging, voting). LA *selects* one model per instance based on learned context-dependent policy.
2. **Advantages of selection**:
   - Computational efficiency: Only run one heuristic, not all 12
   - Interpretability: Know which heuristic was used for each prediction
   - Handles incompatible heuristics: Some may give completely wrong answers in certain conditions; averaging would hurt
3. **Related work**: This is related to "hyper-heuristics" literature. LA provides principled selection mechanism with convergence guarantees.
4. **Fair comparison**: Will include majority vote ensemble as baseline.

Q22: Why not just train one big end-to-end model that handles all conditions?
A:

1. **Data requirements**: End-to-end model needs labeled data covering all conditions. We have sparse, biased labels.
2. **Interpretability**: Black-box model doesn't tell us *why* it works or fails in specific conditions.
3. **Modularity**: Pre-trained heuristics can come from different sources (published methods, domain experts). Don't need to retrain everything.
4. **Adaptation speed**: LA adapts with few labels. Retraining end-to-end model requires many labels and compute.
5. **Empirical question**: This is testable. U-Net baseline represents this approach. If it wins convincingly, that's a valid finding.

Q23: How do you know the heuristic pool has sufficient diversity?
A:

1. **Design rationale**: Three tiers (indices, ML, DL) represent fundamentally different approaches with different inductive biases.
2. **Literature support**: Prior work shows spectral indices and ML have complementary strengths (Schroeder 2020, Gendall 2023).
3. **Empirical validation**: Will analyze which heuristics win in which conditions. If one tier dominates everywhere, others could be removed.
4. **Expandability**: Pool can be extended. If new method emerges (e.g., foundation models), can add as heuristic.

5. **Oracle comparison**: Gap between LA and per-tile oracle indicates room for improvement. Large gap suggests need for more/better heuristics.

Q24: What's the novel scientific contribution beyond engineering?
A:

1. **Novel LA application**: First application of LA to satellite-based environmental monitoring. Extends LA theory to new domain.
2. **Sparse feedback theory**: Formalizing delayed reward propagation with temporal decay. Connects LA to experience replay in RL.
3. **Context-aware GLA**: Practical implementation of Generalized Learning Automata principles with memory-based adaptation.
4. **Empirical science**: Characterizing heuristic complementarity across environmental conditions. This is new knowledge about kelp detection.
5. **Methodology transfer**: Framework applicable to other monitoring problems (seagrass, coral). Generalizable contribution.
6. **Multi-source data fusion framework**: Novel methodology for fusing heterogeneous environmental data sources (satellite imagery, oceanographic data, tidal predictions) with principled handling of resolution mismatches and missing data. This is a standalone methodological contribution.

5.8 Deep Technical Questions (Expect from Dr. Costa - Remote Sensing)

Q25: How do you handle mixed pixels where kelp only partially covers a 10m×10m area?
A:

1. **The mixed pixel problem**: At 10m resolution, a pixel may contain 30% kelp, 70% water. Binary classification forces a decision.
2. **Current approach**: Threshold-based. If spectral signature suggests >50% kelp, classify as kelp. This is standard practice in kelp remote sensing.
3. **Spectral unmixing alternative**: Could use linear spectral unmixing to estimate sub-pixel kelp fraction:
$$\mathbf{r}_{\text{pixel}} = f_{\text{kelp}} \cdot \mathbf{r}_{\text{kelp}} + f_{\text{water}} \cdot \mathbf{r}_{\text{water}} + \epsilon$$
But this requires pure endmember spectra, which vary with conditions.
4. **Why binary is acceptable**: For coastline-scale monitoring, we care about kelp presence/absence at tile level. Sub-pixel fraction is secondary. Also, our evaluation uses IoU which is robust to boundary effects.
5. **Future extension**: Could add fractional kelp cover as soft output from some heuristics.

Q26: Sentinel-2 has known issues with adjacency effects near coastlines. How do you handle this?
A:

1. **The problem**: Bright land adjacent to dark water causes "bleeding" of land signal into water pixels due to atmospheric scattering and sensor PSF.

2. **Magnitude**: Can affect pixels up to 1-2 km from shoreline—exactly where kelp occurs.

3. **Mitigation strategies**:

   - Use Level-2A (atmospherically corrected) which partially addresses this
   - Exclude pixels within 100m of land mask (but lose some kelp)
   - Include "distance to shore" as context feature so LA can learn which heuristics handle adjacency better

4. **Empirical approach**: Some heuristics (especially deep learning) may implicitly learn to handle adjacency. Let the LA discover which work best near shore vs. offshore.

5. **Acknowledgment**: This is a known limitation. Perfect correction would require radiative transfer modeling beyond scope of this thesis.

Q27: How do you validate that your SST/tide/turbidity values are actually correct for a given tile?
A:

1. **SST validation**: Compare NOAA CoralTemp to BC buoy network (Environment Canada) where co-located. Expect RMSE <1°C based on product validation literature.

2. **Tide validation**: CHS predictions are highly accurate (±15 cm for major stations). For interpolated locations, compare to water line position visible in imagery where possible.

3. **Turbidity validation**: Hardest to validate. Can compare Sentinel-3 Kd490 to in-situ Secchi depth measurements from research cruises. Expect correlation >0.7 based on prior BC studies.

4. **Indirect validation**: If context features are wrong, the LA should perform poorly because similar "contexts" would have inconsistent performance. Good LA performance is indirect evidence of reasonable context quality.

5. **Sensitivity analysis**: Will test framework with deliberately perturbed context features to assess robustness.

Q28: What about sun glint? It's a major problem for coastal optical remote sensing.
A:

1. **The problem**: Specular reflection of sunlight off water surface creates bright artifacts that mask kelp signal.

2. **When it occurs**: Primarily when sun angle and viewing geometry align—varies with latitude, time of day, season.

3. **Mitigation in preprocessing**:

   - Sentinel-2 Level-2A includes glint flags in Scene Classification
   - Can filter tiles with severe glint
   - Compute glint probability from sun/sensor geometry

4. **Context-aware handling**: Include solar zenith angle as context feature. LA can learn that certain heuristics (e.g., NIR-based indices) are more affected by glint than others (e.g., red-edge indices).

5. **Temporal compositing**: For extent mapping (not real-time), can composite multiple acquisitions to fill glint-affected areas.

Q29: The 5km resolution of SST is very coarse for 10m kelp detection. How is this useful?
A:

1. **Acknowledged limitation**: Yes, 5km SST cannot capture sub-kilometer thermal gradients (e.g., upwelling zones, river plumes).

2. **Why it's still useful**:

   - Regional SST indicates whether we're in warm (stressful) or cool (favorable) conditions
   - Kelp stress at 18-20°C is a regional phenomenon, not pixel-level
   - SST helps distinguish summer from winter, heatwave years from normal years

3. **Higher resolution alternatives** (potential extensions):

   - MODIS SST: 1km but more gaps due to clouds
   - Landsat thermal: 100m but 16-day revisit
   - Sentinel-3 SLSTR: 1km, better coverage

4. **Empirical test**: Ablation study will remove SST feature. If performance drops significantly, SST is useful despite coarse resolution. If not, can deprioritize.

5. **Key insight**: The framework is robust to imperfect context features. If SST doesn't help, the LA will learn to ignore it (by having similar performance across SST values).

5.9 Deep Technical Questions (Expect from Dr. Stanley - Machine Learning)

Q30: Your k-NN retrieval assumes Euclidean distance in feature space is meaningful. Have you validated this?
A:

1. **Valid concern**: Euclidean distance treats all features equally and assumes isotropic relevance.

2. **Mitigation 1 - Normalization**: Z-score normalization ensures all features have comparable scale. Without this, high-variance features would dominate.

3. **Mitigation 2 - Feature selection**: Ablation studies will identify which features are most informative. Can weight or remove uninformative ones.

4. **Alternative distances**:

   - Mahalanobis distance: Accounts for feature correlations. Requires covariance estimation.
   - Learned metric: Train a Siamese network to learn similarity. More data-hungry.

- Weighted Euclidean: Learn per-feature weights via cross-validation.

5. **Empirical validation**: Compare k-NN retrieval quality using different metrics. If Euclidean works well, no need to complicate.

6. **Fallback**: If k-NN doesn't help (ablation shows context-aware $\approx$ global-only), then distance metric is moot anyway.

Q31: How do you set the blending coefficient α? This seems like a critical hyperparameter.
A:

1. **The role of α**: Controls global vs. local policy influence. $\alpha = 1$ means pure global; $\alpha = 0$ means pure local.

2. **Adaptive α based on confidence**:

$$\alpha(\mathbf{c}) = \frac{1}{1 + \exp(-\beta \cdot (\bar{d} - d_{\text{thresh}}))}$$

   where $\bar{d}$ is mean distance to k neighbors. Far from any memory entry $\Rightarrow$ high α (trust global). Close to many entries $\Rightarrow$ low α (trust local).

3. **Hyperparameters**: β (steepness) and d_{thresh} (transition point) tuned via validation.

4. **Simple alternative**: Fixed $\alpha = 0.5$ as starting point. Let experiments determine if adaptive α helps.

5. **Ablation**: Compare fixed $\alpha \in \{0, 0.3, 0.5, 0.7, 1.0\}$ vs. adaptive. Report sensitivity.

Q32: What prevents the LA from getting stuck in a local optimum where one heuristic dominates?
A:

1. **$\mathbf{L_{R\text{-}I}}$ behavior**: Theoretically converges to optimal action with probability approaching 1 as $\lambda \to 0$. But in non-stationary environments, "optimal" changes.

2. **Context-aware rescue**: Even if global policy converges to one heuristic, local adaptation can still favor others for specific contexts. This is why Component 2 is critical.

3. **Exploration mechanisms** (if needed):

   - ϵ-greedy: With probability ϵ, select random heuristic
   - Softmax temperature: $p_i \propto \exp(q_i/T)$ with temperature T controlling exploration
   - Pursuit algorithms: Maintain estimates of all heuristics, explore under-sampled ones

4. **Non-stationarity helps**: Seasonal changes mean no single heuristic is always optimal. LA must maintain diversity.

5. **Monitoring**: Track entropy of probability distribution $H(\mathbf{p}) = -\sum_i p_i \log p_i$. Alert if $H \to 0$ (premature convergence).

Q33: How do you handle the cold start problem? Initially, you have no memory bank entries.

A:

1. **Phase 1 - Pure exploration**: With empty memory bank, $\alpha \to 1$, so we use global policy only. Global policy starts uniform, so all heuristics get tried.

2. **Bootstrapping options**:
   - Run all 12 heuristics on historical labeled data, populate memory bank with results
   - Use published literature to set initial priors (e.g., "NDVI works well in clear water" from Schroeder 2020)
   - Transfer memory bank from similar geographic region if available

3. **Warm-up period**: First N tiles (e.g., N=1000) use uniform random selection to ensure all heuristics are tried across diverse conditions.

4. **Convergence analysis**: Report how many labeled tiles needed before context-aware adaptation provides benefit over global-only.

Q34: Your reward is IoU, but IoU is undefined when both prediction and ground truth have zero kelp pixels. How do you handle this?

A:

1. **The edge case**: If $|\hat{\mathbf{y}}| = 0$ and $|\mathbf{y}| = 0$, then IoU $= 0/0 =$ undefined.

2. **Interpretation**: Both predicted "no kelp" and ground truth is "no kelp." This is a **correct** prediction!

3. **Solution**: Define IoU for this case as 1.0 (perfect agreement):

$$\text{IoU} = \begin{cases} 1.0 & \text{if } |\hat{\mathbf{y}} \cup \mathbf{y}| = 0 \\ \frac{|\hat{\mathbf{y}} \cap \mathbf{y}|}{|\hat{\mathbf{y}} \cup \mathbf{y}|} & \text{otherwise} \end{cases}$$

4. **Alternative**: Use F1-score with similar handling, or use accuracy for no-kelp tiles.

5. **Class imbalance consideration**: Most tiles may have little/no kelp. Ensure evaluation stratifies by kelp presence to avoid misleading metrics.

Q35: With 29 context features, aren't you at risk of overfitting with limited labeled data?

A:

1. **Valid concern**: High-dimensional context + sparse labels = potential overfitting in k-NN.

2. **k-NN is non-parametric**: Unlike neural networks, k-NN doesn't have parameters to overfit. It just memorizes and retrieves.

3. **Regularization via k**: Larger k means more neighbors averaged, smoother estimates, less overfitting. Will tune $k \in \{3, 5, 10, 20\}$.

4. **Dimensionality reduction** (if needed):
   - PCA: Reduce to top 10 principal components
   - Feature selection: Keep only features with demonstrated predictive value
   - Autoencoder: Learn compressed context representation
5. **Ablation evidence**: If full 29-feature context doesn't outperform reduced feature sets, simplify.
6. **Memory bank size**: With 10,000+ labeled tiles, 29 features is reasonable. Curse of dimensionality is more severe with fewer samples.

Q36: How do you ensure reproducibility given the stochastic nature of LA?
A:

1. **Sources of randomness**: Heuristic selection is sampled from probability distribution. Different runs may select different heuristics.
2. **Seed control**: Fix random seeds for all experiments. Report exact seeds used.
3. **Multiple runs**: Report mean ± std over 5+ random seeds for all metrics.
4. **Deterministic mode**: For deployment, can use argmax selection (always pick highest probability heuristic) instead of sampling. Loses exploration but ensures reproducibility.
5. **Code release**: Will release all code, data splits, and trained models for reproducibility.

5.10 Fundamental/Philosophical Questions

Q37: Why should we believe LA is better than simpler baselines for this problem?
A:

1. **Honest answer**: We don't know yet. This is an empirical research question.
2. **Hypothesis**: LA's theoretical properties (sparse feedback handling, convergence guarantees, context-aware adaptation) make it *plausibly* better for this problem.
3. **Rigorous comparison**: Will compare against:
   - Random selection (lower bound)
   - Best single heuristic (strong baseline)
   - Majority vote ensemble (simpler adaptive approach)
   - Contextual bandit (standard RL baseline)
   - Supervised meta-learner (if enough labeled data)
4. **Negative result is valid**: If LA doesn't outperform baselines, that's a publishable finding. We'd analyze why and what this tells us about the problem structure.
5. **Contribution regardless**: Even if LA ties with baselines, the data fusion framework, sparse label handling mechanism, and heuristic complementarity analysis are standalone contributions.

Q38: Is this really a PhD-level contribution, or is it "just" applying existing methods?

A:

1. **Novel application**: LA has never been applied to satellite-based environmental monitoring. Novel application to important domain is valid contribution.
2. **Methodological extensions**:
   - S-model continuous rewards (extending P-model theory)
   - Memory-based context adaptation (novel GLA implementation)
   - Temporal decay for delayed feedback (new mechanism)
   - Multi-source data fusion with quality flags (methodological contribution)
3. **Empirical science**: Characterizing heuristic performance across environmental conditions is new knowledge.
4. **Practical impact**: A working kelp monitoring system for BC's 26,000 km coastline would be significant.
5. **Multiple papers**: The work supports 5 distinct publications (see contribution section), indicating sufficient depth.

Q39: What if the heuristics themselves are not good enough? The LA can only select; it can't improve a bad heuristic.

A:

1. **Correct observation**: LA is a meta-algorithm. If all heuristics fail for some condition, LA fails too.
2. **Mitigation 1 - Diverse pool**: Three tiers with fundamentally different approaches. Unlikely all fail simultaneously.
3. **Mitigation 2 - Pool expansion**: If systematic failure mode identified, add new heuristic targeting that condition.
4. **Mitigation 3 - Confidence flagging**: Track variance across heuristics. High variance = uncertain. Flag for human review.
5. **Mitigation 4 - Fine-tuning**: Deep learning heuristics can be fine-tuned on new labeled data from failure cases.
6. **Upper bound analysis**: Oracle baseline (per-tile best heuristic) shows achievable performance. Gap between LA and oracle indicates room for better selection. Gap between oracle and 100% indicates need for better heuristics.

Q40: How does this relate to foundation models / large pretrained models that are dominating ML now?

A:

1. **Complementary, not competing**: Foundation models could be added as a heuristic in the pool. LA then decides when to use the expensive foundation model vs. simpler methods.

2. **Cost-aware selection**: Foundation models are computationally expensive. Multi-objective reward could favor simpler heuristics when they suffice, use foundation model only when needed.

3. **Domain adaptation**: Foundation models trained on general imagery may not work well on coastal satellite data without fine-tuning. LA can learn this empirically.

4. **Emerging opportunity**: Segment Anything Model (SAM), CLIP for remote sensing, etc. These could be future heuristics.

5. **Thesis scope**: Focus is on LA framework, not on developing new detection models. Foundation models are potential extension.

5.11 Data Fusion Specific Questions

Q41: The data fusion you describe (satellite + SST + tide + ocean color) seems like a significant contribution on its own. Is this a separate paper?
A:
Yes, absolutely. This is now planned as a core paper in the thesis.

1. **Paper scope**: "Multi-Source Environmental Data Fusion for Coastal Ecosystem Monitoring: A Framework with Principled Missing Data Handling"

2. **Novel contributions**:

   - Systematic methodology for fusing data at 5 different resolutions (10m to 5km)
   - Hierarchical fallback strategy for missing data (temporal → spatial → climatology → imputation)
   - Quality-aware feature vectors that track data provenance
   - Validation framework for multi-source alignment

3. **Generalizability**: The fusion framework applies beyond kelp—any coastal monitoring (seagrass, coral, water quality) faces same challenges.

4. **Target venue**: Remote Sensing of Environment, ISPRS Journal of Photogrammetry, or similar.

5. **Standalone value**: Even if LA doesn't outperform baselines, the data fusion paper stands alone as methodological contribution.

Q42: How do you validate that the fused data actually improves prediction compared to using imagery alone?
A:

1. **Ablation study design**:

   - Baseline: Imagery features only (spectral + texture)
   - +Temporal: Add DOY, days since last observation
   - +Environmental: Add SST, tide, chlorophyll, turbidity
   - +Geographic: Add lat, lon, distance to shore, fetch, depth

- Full: All 29 features

2. **Metrics**: Compare context-aware LA performance with each feature subset.
3. **Feature importance**: Analyze which context features most affect k-NN retrieval and LA performance.
4. **Hypothesis**: Environmental features (SST, tide, turbidity) will provide most benefit because they capture conditions not visible in imagery alone.
5. **Negative result handling**: If imagery-only performs equally well, report this as finding—simpler is better.

Q43: What about uncertainty propagation? Each data source has measurement errors that should propagate to predictions.
A:

1. **Valid point**: SST has ±0.5°C error, tide predictions have ±15cm, etc. These uncertainties compound.
2. **Current approach**: Quality flags indicate data source reliability, but don't quantify exact uncertainty.
3. **Potential extension - Bayesian treatment**:
   - Model each context feature as distribution, not point estimate
   - Propagate uncertainty through k-NN (uncertain neighbors get lower weight)
   - Output prediction with confidence interval
4. **Scope limitation**: Full uncertainty quantification is beyond current thesis scope. Acknowledge as future work.
5. **Practical mitigation**: Quality flags and multiple fallback levels implicitly handle uncertainty by distinguishing reliable vs. unreliable data.

6 Key Equations to Know

$\mathbf{L_{R\text{-}I}}$ Update (P-model):

$$p_i(n+1) = p_i(n) + \lambda[1 - p_i(n)] \qquad \text{(reward for action } i\text{)} \tag{7}$$
$$p_j(n+1) = (1-\lambda)p_j(n), \quad j \neq i \qquad \text{(all other actions)} \tag{8}$$

$\mathbf{L_{R\text{-}I}}$ Update (S-model):

$$p_i(n+1) = p_i(n) + \lambda \cdot R \cdot [1 - p_i(n)] \tag{9}$$
$$p_j(n+1) = p_j(n) - \lambda \cdot R \cdot p_j(n), \quad j \neq i \tag{10}$$

IoU Reward:

$$R = \text{IoU}(\hat{\mathbf{y}}, \mathbf{y}) = \frac{|\hat{\mathbf{y}} \cap \mathbf{y}|}{|\hat{\mathbf{y}} \cup \mathbf{y}|}$$

Local Probability (k-NN):

$$\mathbf{p}_{\text{local}}(\mathbf{c}) = \frac{\sum_{j \in \mathcal{N}_k(\mathbf{c})} w_j \cdot R_j \cdot \mathbf{p}_j}{\sum_{j \in \mathcal{N}_k(\mathbf{c})} w_j \cdot R_j}$$

Distance Weights:

$$w_j = \exp\left(-\frac{\|\mathbf{c} - \mathbf{c}_j\|_2^2}{\sigma^2}\right)$$

Global-Local Blending:

$$\mathbf{p}_{\text{final}}(\mathbf{c}) = \alpha(\mathbf{c}) \cdot \mathbf{p}_{\text{global}} + (1 - \alpha(\mathbf{c})) \cdot \mathbf{p}_{\text{local}}(\mathbf{c})$$

Temporal Decay:

$$R_{\text{effective}} = R \cdot \gamma^{\delta}, \quad \delta = t_{\text{label}} - t_{\text{prediction}}$$

Optimality Criterion:

$$\lim_{n \to \infty} E[M(n)] = c_\ell \quad \text{where } c_\ell = \min_i c_i$$

7 Key References to Cite

Kelp Ecology & BC Context:

- Starko et al. (2024): BC kelp decline patterns, 11 subregions
- Mora-Soto et al. (2024): 4 environmental clusters in Salish Sea
- Schroeder et al. (2020): Kelp detection methods, image quality scoring

Learning Automata:

- Narendra & Thathachar (1989): Foundational LA textbook
- Thathachar & Sastry (2002): Modern LA theory
- Bisong & Oommen (2019–2021): My prior OMA publications

Remote Sensing Methods:

- Gendall et al. (2023): Multi-satellite kelp mapping
- Timmer et al. (2024): Tides, currents, morphology for kelp detection
- Cavanaugh et al. (2021): UAV kelp mapping

Machine Learning:

- Drake et al. (2020): Hyper-heuristics survey
- Savargiv et al. (2021): LA for Random Forest optimization
- Sutton & Barto (2018): Reinforcement learning textbook

8 The Data Fusion Paper - A Core Thesis Contribution

This section elaborates on the multi-source data fusion methodology as a standalone, significant contribution warranting its own publication.

8.1 Why Data Fusion is a Core Contribution

The context-aware adaptation mechanism (Component 2) requires fusing data from **5+ heterogeneous sources** with vastly different characteristics:

Source	Spatial Res.	Temporal Res.	Coverage	Primary Challenge
Sentinel-2 imagery	10m	5 days	Global	Cloud gaps
NOAA CoralTemp SST	5 km	Daily	Global	Coarse resolution
Sentinel-3 OLCI	300m	1-2 days	Global	Coastal contamination
CHS Tide Predictions	Station-based	6-minute	BC only	Sparse stations
CHS Bathymetry	100m	Static	BC only	Survey gaps

The resolution mismatch problem: A 10m $\times$ 10m satellite pixel must be associated with environmental context derived from sources ranging from 5 km (SST) to point measurements (tide stations). This is a **500$\times$ to ∞ resolution ratio**.

The missing data problem: Ocean color products have 30-50% missing data due to clouds. How do you build a context vector when half the features might be unavailable?

These are not trivial preprocessing steps—they are methodological contributions.

8.2 Proposed Paper: Multi-Source Environmental Data Fusion

Title: "Multi-Source Environmental Data Fusion for Coastal Ecosystem Monitoring: A Framework with Principled Missing Data Handling"

Target Venues:

- Remote Sensing of Environment (IF: 13.5)
- ISPRS Journal of Photogrammetry and Remote Sensing (IF: 12.7)
- International Journal of Applied Earth Observation and Geoinformation (IF: 7.5)

Paper Structure:

1. **Introduction**: The challenge of coastal monitoring with multi-source data
2. **Related Work**: Review of data fusion methods in environmental remote sensing
3. **Methodology**:
   - Spatial alignment strategies (bilinear interpolation, nearest station, IDW)
   - Temporal alignment (exact match, windowed search, climatology fallback)
   - Hierarchical missing data handling (4-level cascade)
   - Quality-aware feature vectors with provenance tracking
4. **Case Study**: Application to BC kelp monitoring with 5 data sources

5. **Validation**:
   - Comparison of alignment strategies
   - Sensitivity to missing data rates
   - Impact on downstream prediction accuracy
6. **Discussion**: Generalizability to other coastal ecosystems
7. **Code/Data Release**: Open-source fusion toolkit

8.3 Novel Contributions of the Data Fusion Paper

Contribution 1: Systematic Resolution Harmonization

No existing paper provides a complete methodology for harmonizing 10m satellite imagery with 5km oceanographic products. Our approach:

- Bilinear interpolation for gridded products (SST, Chl)
- Nearest-station + temporal interpolation for point data (tide)
- Precomputed rasters for static layers (bathymetry, fetch)
- Validation of interpolation accuracy against independent measurements

Contribution 2: Hierarchical Missing Data Cascade

A principled 4-level fallback strategy:

1. Exact temporal match
2. Temporal window search (±N days)
3. Climatological fallback (seasonal average)
4. Imputation with quality flag

This is more principled than ad-hoc gap-filling and more practical than complex imputation models.

Contribution 3: Quality-Aware Feature Vectors

Augmented context vectors that track data provenance:

$$\mathbf{c}^{\text{aug}} = [\mathbf{c}, q_1, q_2, \ldots, q_m]$$

where $q_i \in \{0, 1, 2, 3\}$ indicates data quality level. This enables downstream models to weight features by reliability.

Contribution 4: Validation Framework

Methods to validate fusion quality:

- Cross-validation against withheld in-situ measurements
- Sensitivity analysis with artificially increased missing data
- Ablation studies showing which data sources provide most value

Contribution 5: Generalizable Toolkit

Open-source Python package for coastal data fusion:

- Modular design for adding new data sources
- Configuration-driven (YAML) for different study regions
- Integration with Google Earth Engine for scalability
- Documentation and tutorials

8.4 Why This Paper Stands Alone

1. **Independent of LA**: The fusion methodology is useful regardless of downstream model. Could be used with any ML approach.
2. **Addresses real gap**: Coastal remote sensing literature often glosses over data fusion as "preprocessing." We make it explicit and rigorous.
3. **Immediate practical value**: Any researcher working with coastal satellite data + oceanographic context can use this framework.
4. **Replicable**: Unlike methods requiring proprietary data, our approach uses freely available data (Sentinel, NOAA, CHS).
5. **Extensible**: Framework can incorporate new data sources (e.g., Landsat thermal, GOES SST, wave buoys).

8.5 Connection to Thesis Narrative

The data fusion paper supports the overall thesis in multiple ways:

Thesis Component	How Data Fusion Supports It
RQ1 (Adaptation)	Context features enable site-specific adaptation
RQ2 (Heterogeneity)	Environmental context distinguishes coastal environments
RQ3 (Sparse Labels)	Quality flags inform reward weighting
RQ4 (TEK)	Framework can incorporate TEK-derived features

Key message for committee: The data fusion work is not peripheral preprocessing—it is a **core methodological contribution** that enables the entire context-aware adaptation mechanism and warrants publication in a top remote sensing venue.

9 Final Checklist Before Presentation

- ☐ Know all key statistics (26,000 km coastline, >70% decline, 18–20°C threshold)
- ☐ Can explain $L_{R\text{-}I}$ update equations from memory
- ☐ Can walk through methodology flow for one tile
- ☐ Know differences between P-model, Q-model, S-model
- ☐ Can explain why LA over standard RL
- ☐ Know all 12 heuristics in the pool

- ☐ Can explain context-aware adaptation mechanism
- ☐ Can explain temporal decay and why it's needed
- ☐ Know evaluation protocols (A, B, C) and what each tests
- ☐ Can defend timeline and acknowledge risks
- ☐ Have backup answers for challenging questions
- ☐ Know key references by author name
- ☐ **Can articulate why data fusion is a core paper contribution**
- ☐ **Can explain the 5-paper publication plan**